

Project: EV Adoption Analysis

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In this report, it presents the project motivation, data sources, integrated data model, visualization and analysis, conclusions, future work, and references.

Motivation

Nowadays, there are more vehicle options for consumers to choose from, including gasoline cars, hybrid cars, and electric vehicles. As electric vehicles become a more popular choice in the US local market, I am interested in understanding why EV adoption is getting higher, especially in some states, with EV being the first choice of vehicle purchase. Personally, I purchased an EV, and it is because of the price. Therefore, this increases my interest in whether cost-related factors, such as income and gasoline price, could impact the EV adoption rates.

The project is exploring the question of what factors may influence electric vehicle adoption across different U.S. states. As we can see, the market has shift giantly from traditional gas cars to electric vehicles, this question becomes much more interesting. Consumers' decisions to purchase an EV may be shaped by affordability, access to charging stations, local gas prices, education level, and age patterns. By analyzing these factors together, this project aims to better understand what factors may support or limit EV adoption within the United States.

Data Source

In this project, 5 datasets were collected and merged to analyze the factors that may influence the EV adoption throughout the US market. These datasets include state-level information in 2023 about EV registration, charging ports, socioeconomic factors, gas prices, and state identifiers. Since some datasets, such as the American Community Survey (ACS), require time to be finalized and publicly released, the 2023 data is the most recent available complete dataset. Therefore, all datasets used in this project were standardized and unified to the year 2023 to ensure consistency and comparability across variables.

Alternative Fueling Station Counts by State (Alternative Fuels Data Center, 2023)

<https://afdc.energy.gov/files/docs/historical-station-counts.xlsx?year=2023>

This dataset was collected to provide one of the non-demographic factors, which is the availability of EV charging stations throughout the US states, and serves as an explanatory variable in the analysis report. This dataset provides the count of alternative fueling stations, including EV charging stations and HPEV charging stations, at the state level. This data is used to analyze how infrastructure accessibility may influence the adoption rate of electric vehicles.

The data was extracted using the `pd.read_excel()` function from the `pandas` library, since the data was directly in Excel format from the AFDC websites. Since the relevant data was stored in the sheet labeled "2023," the extraction process specified the sheet name accordingly. Additionally, parameters such as `skiprows=3` and `skipfooter=6` were applied to remove unnecessary header and footer content, ensuring that only the relevant tabular data was captured. The dataframe was then saved as a CSV file for further processing and integration. The dataset contains approximately 51 sets of observations, corresponding to each US state, and a row of total counts.

Some challenges were encountered in scraping the data, in the Excel file, that were some extra header and footer rows, and another one was an unexpected structure where each state's data spanned two rows, with the second row mostly containing missing values. This was resolved by using parameters `skiprows` and `skipfooter` in `pd.read_excel()`, and applying a loop with a step size of 2 in the `range()` function to correctly extract and consolidate data for each state.

American Community Survey (U.S. Census Bureau, 2023)

<https://www.census.gov/data/developers/data-sets/acs-1year.2023.html>

This dataset was collected to provide several key socioeconomic and demographic variables, including median household income, education level, population, and age distribution. These variables were used as independent variables to analyze the relationship between those factors and EV adoption rates at a state level.

This dataset was obtained through the US Census Bureau API by constructing a request URL that specifies relevant variable codes and retrieves data for all states. The `requests` library was used to access the API, and the returned JSON type of the data was converted into a structured dataframe using `pd.DataFrame()` function from `pandas`. The extracted dataset contains approximately 50 observations, corresponding to each US state.

A key challenge in extracting and cleaning the dataset was that variable names provided by the Census API are encoded (e.g., B01003_001E), which may be difficult to interpret directly. To address this issue, the official Census variable documentation was referenced to correctly identify and select the appropriate variables for income, education, and age groups.

Gas Prices in Every U.S. State (Visual Capitalist, 2023)

<https://www.visualcapitalist.com/mapped-gas-prices-in-every-u-s-state/>

This dataset was collected to provide a non-demographic factor that may impact the EV adoption rate, which is the gasoline prices throughout the US. Including this dataset for this analysis could extend the scope of the project beyond demographic variables, as the original plan placed a heavier emphasis on age-related factors. By including the gas price as a socioeconomic variable, the analysis could better capture cost-related influences on EV adoption behavior.

This dataset was used with the same web scraping technique as the `Two-Letter State Abbreviation` dataset. The scraping process used `requests` and `BeautifulSoup` libraries in Python. The `requests` library makes a request to the web page, then retrieves the HTML content. `BeautifulSoup` was used to parse the HTML structure. The `<td>` tags were extracted by using `find_all("td")` since the target raw data was stored in table content in the original web page. The raw data extracted contains a total of 51 pairs of table cell values, and have stored in a `.txt` file in the `data/raw` folder.

One change from the original project plan was the addition of this dataset to provide a more balanced analysis between demographic and non-demographic factors. A big challenge encountered during this dataset collection was the instability of web scraping, as changes to the website structure or system conditions could result in errors when running the scraping function Python file. This issue was addressed by re-running the script or revalidating the data extraction process when necessary.

Two-Letter State Abbreviation (Federal Aviation Administration)

https://www.faa.gov/air_traffic/publications/atpubs/cnt_html/appendix_a.html

This dataset was collected to match each US state name with its corresponding two-letter abbreviation. The purpose of this dataset is to provide the information that later visualization analysis, which a choropleth map requires, is the correct state abbreviations to display state-level geographic information data.

This data was collected through a web scraping technique by using `requests` and `BeautifulSoup` libraries in Python. Using `requests.get()` to request access to the website and retrieve the HTML content. Then, use `BeautifulSoup` to parse the HTML structure. In the original webpage, the state names and abbreviations are stored inside table cells; the `<td>` tags were extracted by using `find_all("td")`. The raw data extracted contains a total of 57 sets of table cell values, and have stored in a `.txt` file in the `data/raw` folder.

The major challenge encountered in this process was that the raw table content was not well-structured, making it difficult to directly convert into a dataframe. Therefore, the raw data was first stored in a text file before the cleaning process and converted into a structured dataframe format.

Vehicle Registration Counts by State (Alternative Fuels Data Center, 2023)

<https://afdc.energy.gov/vehicle-registration?year=2023>

This dataset was collected to calculate and measure the EV adoption rate across the US states and will serve as the primary source for the dependent variable in this analysis. The data provides the state-level information on vehicle registration counts by fuel type. The purpose of this dataset is to form the foundation of this analysis by providing the dependent variables. Compared to the original project plan, the dataset was specifically constrained to the year 2023 to ensure consistency with other data sources.

This dataset was collected using the `pandas` library, specifically the `pd.read_html()` function to extract the tabular data from the webpage. The function will retrieve all the tables on the target webpage, and then the relevant table that contains the vehicle registration data will be stored in a pandas dataframe. The raw data will be stored in a CSV file for further cleaning process and analysis. The dataset includes approximately 50 observations, corresponding to each US state, and also a row of total counts of all states.

Integrated Data Model

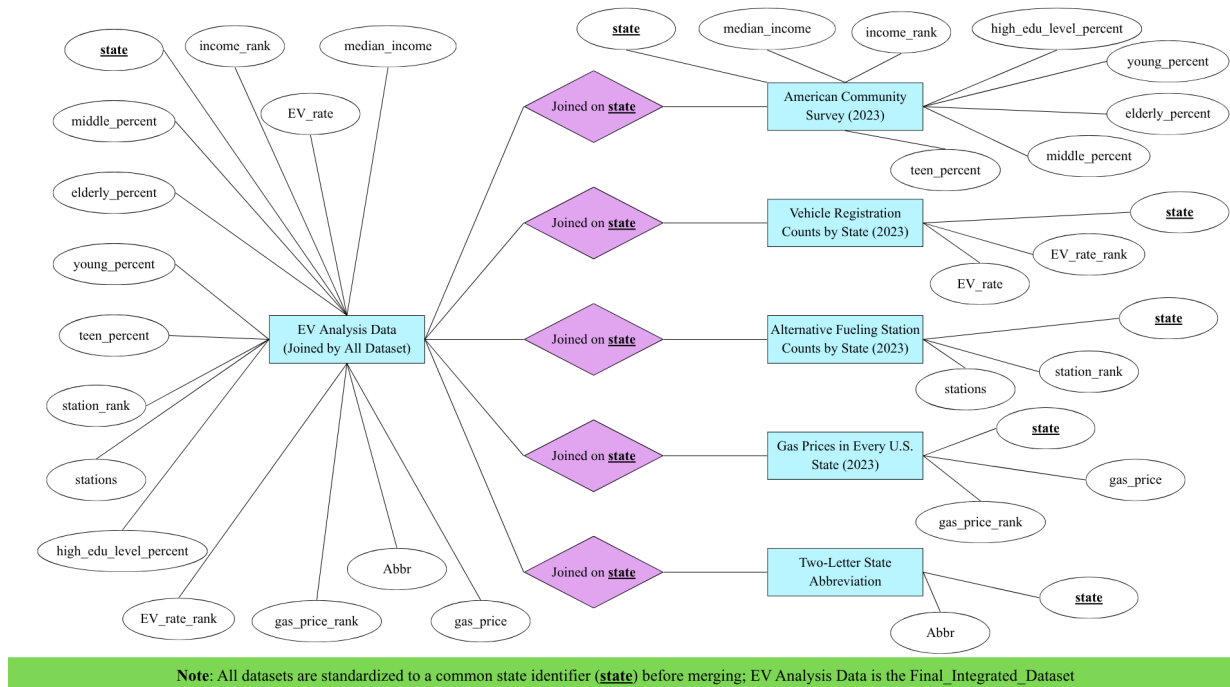
In this project, the integrated data model was developed by merging five distinct datasets into a single standardized dataframe using `pandas` functions. To ensure consistency throughout all data sources, I cleaned and standardized the datasets, using the state as the common key and as the primary key in the model. This standardization enables accurate merging and alignment of variables from different dataframes into the final dataset, as the CSV file called "FINAL_DATA" in the data folder.

The model is centered on the final integrated dataset, which merged information on EV adoption, gas prices, charging station counts, socioeconomic characteristics, demographic variables, and the state abbreviation. Each data source has joined with other datasets based on the same primary key, the state. The EV registration dataset provides the main outcome variable, EV adoption rate, while the ACS, gas price, and charging station datasets provide

explanatory variables used for analysis. The state abbreviation dataset supports standardization and visualization, especially for the choropleth map.

Informal entity-relationship diagram

This diagram shows that all five original datasets are merged into the final integrated dataset through the shared primary key, which is the state. This model allows the analysis to compare the EV adoption rates with several potential explanatory factors within one dataframe.



Please refer to the results folder if the diagram is hard to read.

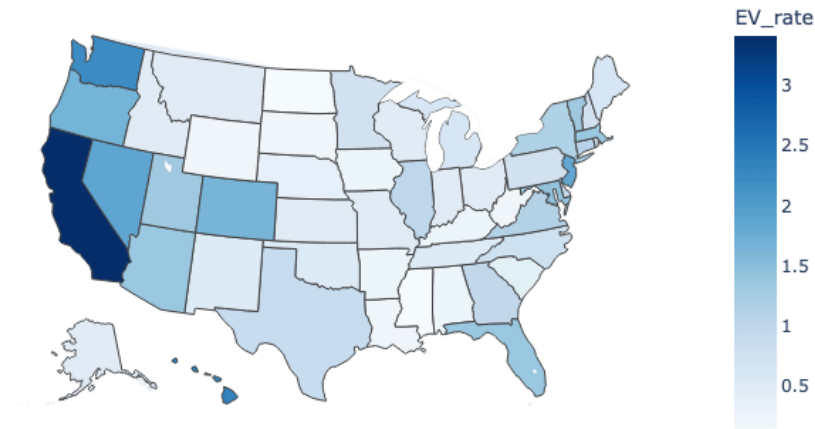
Visualization Analysis

To effectively support the analysis of the project, multiple types of visualizations were used to display different aspects of the data. The interactive choropleth map was created to show the overall distribution of EV adoption rates across the US for each state. Linear regression plots were generated to examine the relationship between individual factors that were identified in an earlier stage and EV adoption rates. A distribution plot of EV adoption rates was also included to show the overall spread and variation across states. Also, a statistical summary table was included to show the spread and variation. With all those visualizations, they provide both an exploratory and analytical perspective on the factors influencing EV adoption.

Note: All plots have been saved in the **results/plots** folder. Please check that folder for a clearer version of plots, and the HTML file for the interactive choropleth map.

EV Adoption Rate Choropleth Map

EV Adoption Rate by State

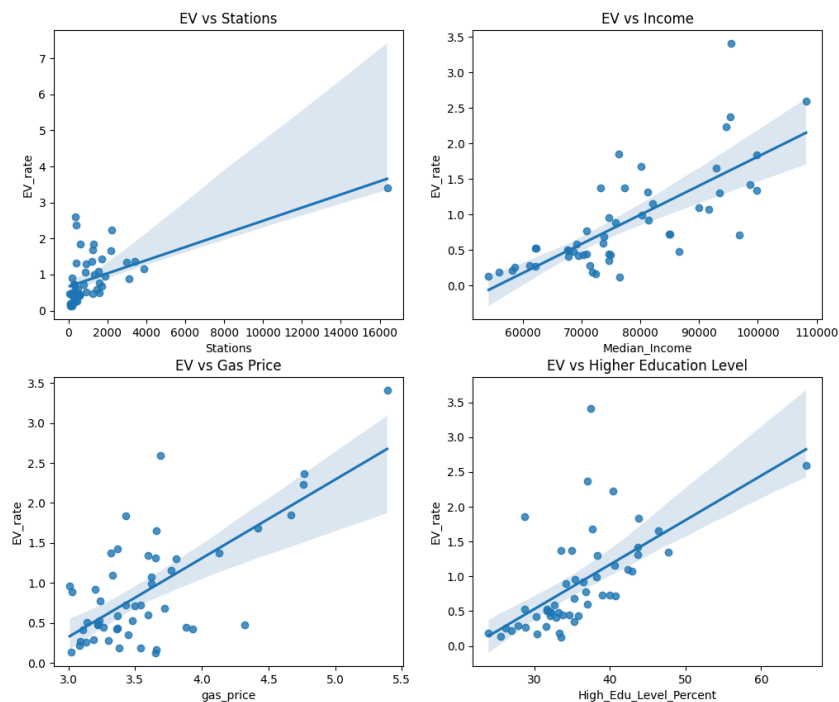


In the project, a choropleth map visualization was used to examine the geographic distribution of EV adoption rate throughout the US. The figure was created using [plotly.express](#) library, which was guided by the [plotly](#) parameter library instructions. The state abbreviations were mapped to geographic locations, and the EV adoption rate was encoded using a continuous color scale. The lighter the blue in the scale, the lower the EV adoption rate in the relevant state. Additionally, variables including EV adoption rank, station rank, gas price rank, and income rank were incorporated into the hover feature to provide more detailed information for each state. For example, when hovering over California state, we can see that the gas price and EV stations ranking is number 1. The visualization was exported as both an interactive HTML file and a static image.

The choropleth map shows clear regional disparities in EV adoption. States on the West Coast, particularly California, show the highest adoption rates with a dark blue scale, while many states in the Midwest and southern regions display relatively lower levels. Northeastern states also demonstrate moderate to high adoption. These patterns suggest that EV adoption is geographically uneven and may be influenced by regional differences in infrastructure, economic conditions, and policy environments.

Non-age Factor Linear Regression

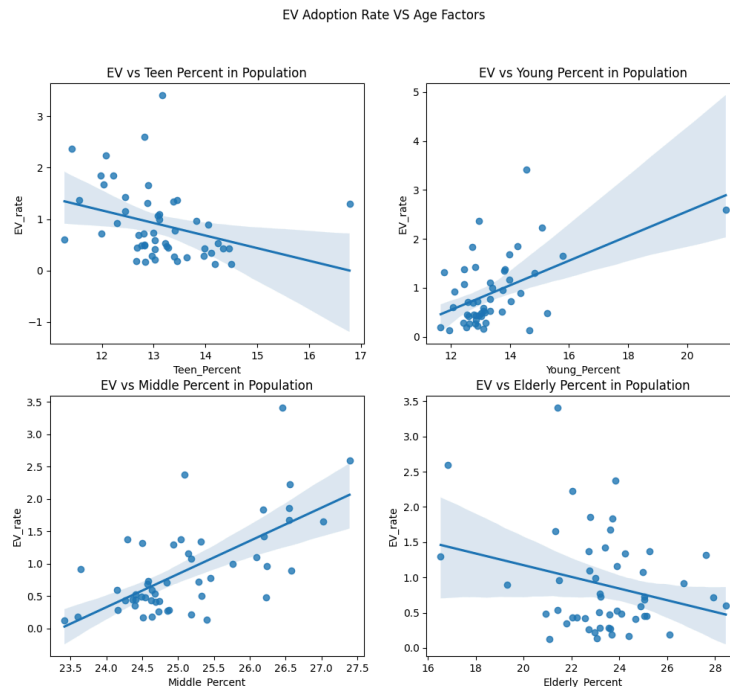
EV Adoption Rate VS Non-age Factors



This visualization is used to analyze correlation and regression relationships, which are used to examine the relationship between non-age factors and EV adoptions. Linear regression plots were created using the `seaborn` library, particularly the `sns.regplot()` function, to visualize the association between EV adoption rate and four explanatory variables, which are charging station counts, median income, gas price, and higher education level percentage in the population. The figure contains 4 subplots that are arranged in a two-by-two layout, each showing a scatter plot and a fitted linear regression, where the x-axis represents each independent variable, and the y-axis represents the EV adoption rate, which is the dependent variable. The regression model and the confidence interval show the direction and the strength of the relationship.

The linear regression plots suggest that there is a positive relationship between the EV adoption rate and all four non-age factors. Specifically, the charging station count variable shows a moderate positive trend due to an outstanding outlier, suggesting that infrastructure availability is a driver of EV adoption but not in an essential role. The other three factors, including median income, gas prices, and education level, demonstrate clear upward relationships, indicating that higher socioeconomic capacity, gas price behavior, and education attainment are associated with greater adoption. Overall, these findings from the visualizations suggest that socioeconomic and infrastructure factors play a significant role in influencing EV adoption across states.

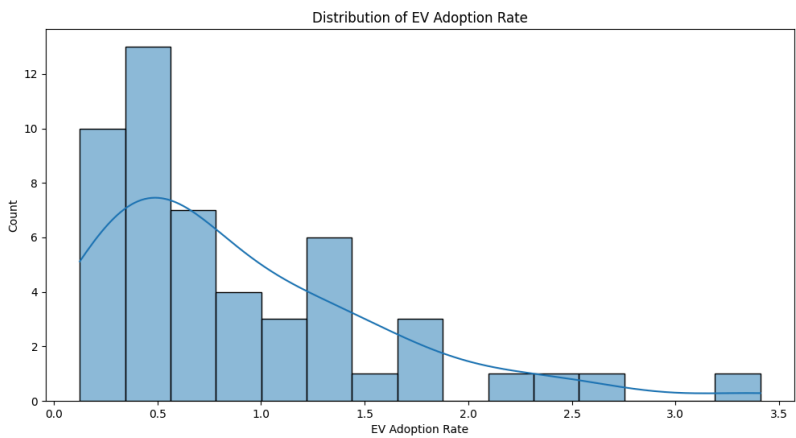
Age Factor Linear Regression



A similar visualization technique was used with the non-age factor linear regression visualization. This visualization also conducted a correlation and linear regression analysis, which was used to examine the relationship between age-related demographic factors and EV adoption rates. The visualization used the `sns.regplot()` function from the `seaborn` library to plot four scatter plots with a fitted linear regression line, which were created to visualize the relationship between EV adoption rate and the population percentage of teens, young, middle-aged, and elderly. The figure is created by organizing into a two-by-two layout, where each subplot represents one age group, the x-axis represents the population percentage of each age group, which is the independent variable, and the y-axis represents the dependent variable, which is the EV adoption rate. The regression line and confidence interval provide a visual indication of the direction and strength of the relationship.

This visualization of the linear regression plots suggests that among all age groups, the young and middle-aged populations have a positive correlation with EV adoption, which indicates that states with higher proportions of these groups tend to have higher adoption rates. In contrast, the elderly and teen population shows a negative relationship, indicating that states with larger elderly and teen populations may have lower EV adoption. Overall, these findings suggest that age structure plays a role in EV adoption, with younger and working-age populations more likely to adopt EVs compared to older or teen populations.

EV Rate Distribution



This visualization was created to develop a descriptive statistical analysis, which was conducted to examine the overall distribution of EV adoption rates among all states. Using the `sns.histplot()` function from the `seaborn` library, and was guided by the seaborn use library. This created a histogram with a kernel density estimate, which shows the frequency distribution of the EV adoption rate. This plot uses several bins to show the spread of values, while the KDE curve feature could provide a smoothed representation of the distribution, where the x-axis represents the EV adoption rate, and the y-axis represents the number of stats within each bin range. This visualization helps illustrate how EV adoption levels are distributed across all observations.

This distribution plot displays that EV adoption is right-skewed, with most states clustered at lower adoption levels and a smaller number of states having higher rates. The majority of states fall within a relatively low range of EV adoption, while a few states appear as higher-value outliers. This indicates that EV adoption is uneven across states and concentrated in a limited number of regions, highlighting disparities in adoption levels nationwide.

Statistical Table

Factor	Correlation (r-value)	p-value
Stations	0.604	0.0000026498314419175527
Median_Income	0.745	3.737159837141388e-10
gas_price	0.697	1.3278304189861084e-8
High_Edu_Level_Percent	0.624	9.742994607441905e-7
Teen_Percent	-0.319	0.022616273896135748
Young_Percent	0.513	0.00011686900916750545
Middle_Percent	0.659	1.425689017409961e-7
Elderly_Percent	-0.262	0.06353211922797329

The final visualization was conducted using Pearson correlation to quantify the relationship between all the explanatory variables and EV adoption rate. The `scipy.stats.pearsonr()` function was used to calculate both the correlation coefficient, which is known as the r-value, and the p-value for each variable. The r-value measures the strength and direction of a linear relationship, ranging from -1 to 1, where positive values indicate a positive relationship and negative values indicate a negative relationship. The p-value evaluates the statistical significance of the relationship, with smaller values typically less than

the alpha value of 0.05, indicating that the observed correlation is unlikely to be due to random chance. By using the `plotly.graph_objects`, the calculation results are converted to a dataframe, then visualized as a table with columns for each factor, its corresponding r-value, and p-value.

The statistical results it shows that most factors have a statistically significant relationship with EV adoption rate, as their p-values are significantly below 0.05. In particular, median income ($r = 0.745$) has the strongest positive correlation, as its r-value is closest to 1, which indicates a strong linear relationship with EV adoption rate. Other factors, such as gas price, charging station counts, and higher education level ratio, also demonstrate meaningful positive correlations with low-level p-values (< 0.05), supporting their significance. Among age-related variables, young and middle population percentages show moderate positive relationships. In contrast, the elderly population has a p-value above 0.05, indicating that its negative correlation is not statistically significant. The teen population has a p-value around 0.02, which is close to the 0.05 alpha threshold, suggesting that its negative relationship is weakly significant. Overall, these results indicate that socioeconomic and infrastructure factors have stronger and more reliable relationships with EV adoption compared to age-related factors.

Conclusion

This project finds that the EV adoption rate in the US is primarily impacted by socioeconomic and informational factors. Explanatory factors like median household income, gas prices, charging station counts, and higher education level population ratio show a strong and statistically significant positive relationship with EV Adoption, while the age-related factors have a weaker effect. Especially, income shows as the strongest correlation, indicating that affordability plays a key role in EV adoption decisions. This suggests that improving infrastructure accessibility and economic conditions can significantly support the growth of EV adoption, which the insight could help businesses better understand which local market to focus resources on to invest in electric vehicle sales.

Future Work

One major limitation of this project is that the datasets used are mostly from 2023, since that is the most recent available complete dataset. Due to the delays in data collection and publication, especially for sources like the ACS that hold a massive amount of data, which need a longer time to process and publish, the analysis may have a bit of hysteresis. However, external factors like gas prices have changed significantly in recent months, which may affect the accuracy of the analysis. In future work, the analysis dataset should use newer and up-to-date datasets, as the datasets could become available to better reflect current market conditions.

Also, this project focuses only on a limited number of variables, and there may be other, more significant and important factors that may affect EV adoption has not been included, such as environmental policies. One more part that can improve is addressing potential confounding variables. Future analysis could apply more advanced statistical methods to better isolate the impact of each factor and reduce the influence of compounded effects between variables.

References

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